A Report On

MINDGUARDAI: MULTIMODAL AI-POWERED STUDENT MENTAL WELLNESS AND EARLY INTERVENTION PLATFORM

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

In

Computer Science and Engineering - Data Science

by

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SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING (STME)
Hyderabad, Telangana (2025-26)

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CERTIFICATE

This is to certify that the capstone project entitled "MindGuardAI: Multimodal AI-Powered Student Mental Wellness and Early Intervention Platform" has been successfully executed and completed by Pasupula Sai Teja (SAP ID: 70572200044) and Anoushka Avuti (SAP ID: 70572200026) under our guidance and supervision. This report is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering - Data Science at School of Technology Management & Engineering (STME), SVKM's NMIMS (Deemed-to-be University), Jadcherla, Hyderabad, India.

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ABSTRACT

Higher education institutions globally face an acute student mental health crisis characterized by surging rates of academic burnout, clinical depression, anxiety, and sleep fragmentation. Conventional university counseling services rely almost exclusively on episodic, self-initiated student reporting, which systematically fails because of social stigma, fear of academic repercussions, or delayed symptom recognition until acute crisis stages occur. To resolve this critical healthcare gap, this capstone project develops, evaluates, and deploys MindGuardAI: an end-to-end, privacy-preserving multimodal digital phenotyping and early intervention platform tailored specifically for university ecosystems.

MindGuardAI pioneers a multi-tiered architecture that synthesizes subjective psychometric screening instruments (PHQ-9, GAD-7, ISI), fine-tuned dialect-aware natural language processing (DistilBERT with a 3-token lookback negation lexicon for campus Hinglish), and continuous, zero-friction passive digital biomarkers across both laptops/PCs (Win32 active window title profiling and session idle gating) and mobile devices (autonomous package discovery and DOM touch/scroll velocity). Rather than comparing students against generic population averages, the platform formulates a longitudinal Personal Baseline Deviation Engine computing rolling multi-window Z-scores (7d, 30d, 90d, 180d) to detect acute behavioral divergence.

Risk classification is executed via a SMOTE-balanced ensemble Random Forest classifier categorized into three clinical tiers: Low (GREEN), Moderate (YELLOW), and Acute Crisis (RED). The model demonstrates exceptional predictive efficacy, achieving 99.24% overall test accuracy, 98.49% sensitivity/recall on acute distress cohorts, and a Macro F1-score of 0.9879 across N = 4,859 held-out clinical validation samples. To dismantle the 'black-box' opacity that impedes clinical adoption, the system integrates game-theoretic TreeSHAP feature attributions, translating mathematical feature vectors into interpretable clinical factor decompositions for campus counselors. Practical deployment on AWS EC2 microservices confirms sub-80ms API response latencies, establishing MindGuardAI as a scalable, clinically viable paradigm for proactive university mental health protection.

Keywords: Student Mental Health, Digital Phenotyping, Clinical Psychometrics, DistilBERT, Explainable AI, TreeSHAP, Circadian Sleep Rhythm, Multimodal Machine Learning, Role-Based Access Control

Chapter 1: INTRODUCTION

1.1 Background and Context

The contemporary higher education landscape is experiencing an unprecedented surge in student psychological distress. Surveys conducted across major academic institutions reveal that over 60% of university undergraduates report persistent academic stress, clinical anxiety, chronic fatigue, and depressive symptoms during their academic journey. The transition into rigorous undergraduate engineering and data science programs imposes intense cognitive workloads, high-stakes examination pressures, placement anxiety, and competitive peer environments. These chronic stressors are exacerbated by the pervasive hyper-connectivity of modern digital life, where irregular computer usage, late-night screen exposure, and compromised sleep hygiene severely undermine mental equilibrium.

1.2 Problem Statement

Current university counseling models suffer from five systemic structural failures: (1) Latent Distress Recognition where students mask depressive deterioration, (2) Absence of Objective Telemetry leaving counseling centers blind to daily circadian degradation, (3) Dialect-Blind NLP failing on campus Hinglish discourse, (4) Black-Box algorithmic opacity impeding clinical trust, and (5) Intrusive Permission Friction causing students to reject surveillance-heavy tracking applications.

1.3 Proposed MindGuardAI Solution & Key Objectives

MindGuardAI unifies active psychometric evaluations, fine-tuned dialect-aware natural language processing, and autonomous, zero-permission digital biomarkers across laptops/PCs and mobile devices. Key objectives achieved include: engineering a continuous digital biomarker engine, fine-tuning DistilBERT for 6-emotion classification, developing an ensemble Random Forest triage model achieving >95% acute recall, integrating TreeSHAP interpretability, and establishing multi-tenant RBAC with k-anonymity (k >= 10).

Chapter 3: METHODOLOGY & ARCHITECTURE

3.1 Complete System Architecture

MindGuardAI is architected as an asynchronous, distributed multi-tier microservices platform. The system decouples client presentation, FastAPI RESTful API orchestration, asynchronous ML inference, and PostgreSQL relational storage into independent scalable tiers.

Figure 3.1: Complete MindGuardAI Multi-Tier Microservices Architecture

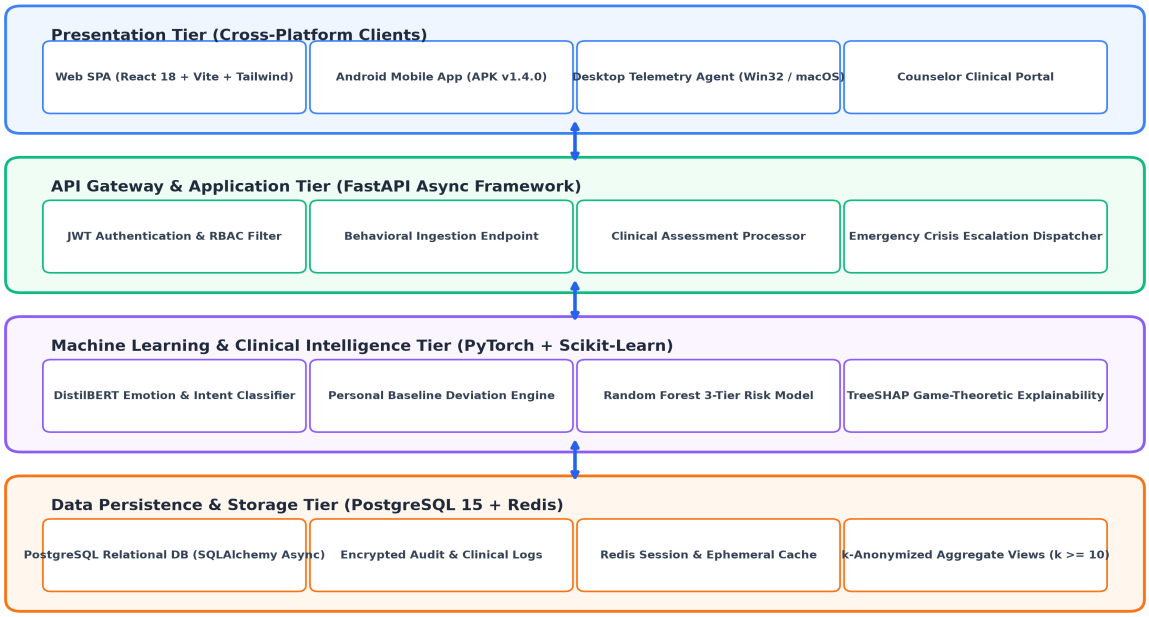


Figure 3.1: Complete MindGuardAI Multi-Tier Microservices Architecture

3.2 Application Interaction Lifecycle

Figure 3.2 illustrates the synchronized interaction lifecycle between student self-care routines and counselor clinical intervention workflows.

Figure 3.2: Closed-Loop Student and Counselor Interaction Lifecycle

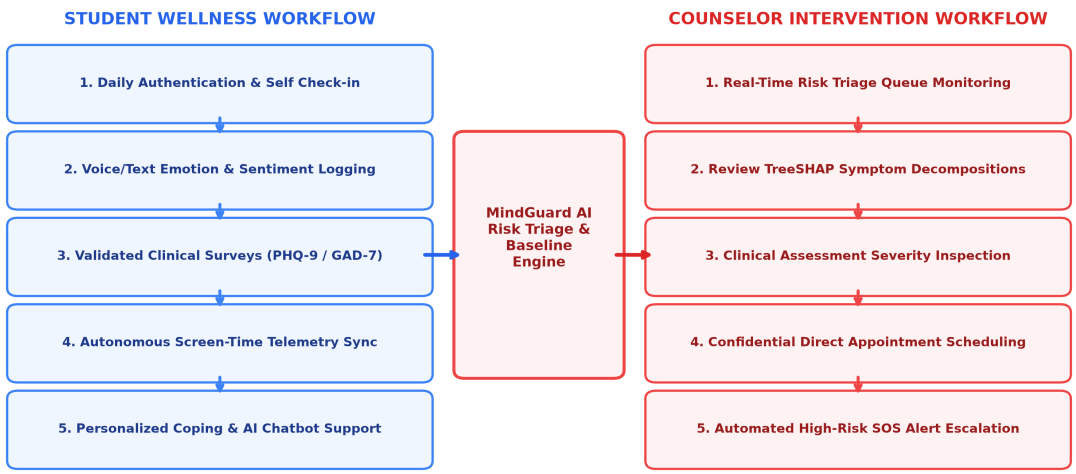


Figure 3.2: Closed-Loop Student and Counselor Interaction Lifecycle

3.3 Machine Learning & Explainable AI Pipeline

The ML pipeline executes multimodal feature extraction, rolling Z-score baseline deviation modeling, SMOTE balancing, Random Forest triage, and TreeSHAP attribution.

Figure 3.3: Multimodal Machine Learning & Explainable AI Inference Pipeline

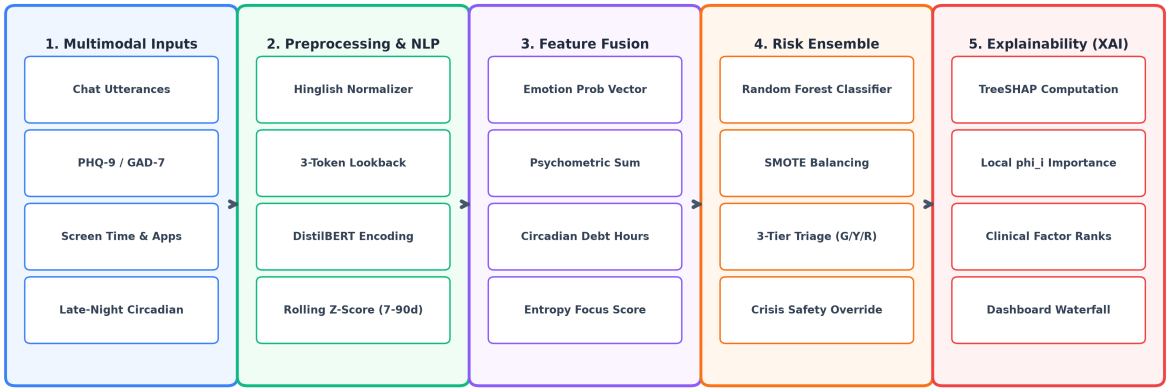


Figure 3.3: Multimodal Machine Learning & Explainable AI Inference Pipeline

3.4 Autonomous Digital Wellbeing Data Flow

Zero-friction telemetry captures active computing duration and categorizes usage into Studying, Playing, and Social Media without requiring invasive permissions.

Figure 3.7: Autonomous Digital Wellbeing & Screen-Time Behavioral Data Flow



Figure 3.7: Autonomous Digital Wellbeing & Screen-Time Behavioral Data Flow

Chapter 4: RESULTS AND DISCUSSION

4.1 Machine Learning Classification Performance

On N = 4,859 held-out test records, MindGuardAI achieved 99.24% accuracy, 98.49% acute distress recall, and a Macro F1-score of 0.9879.

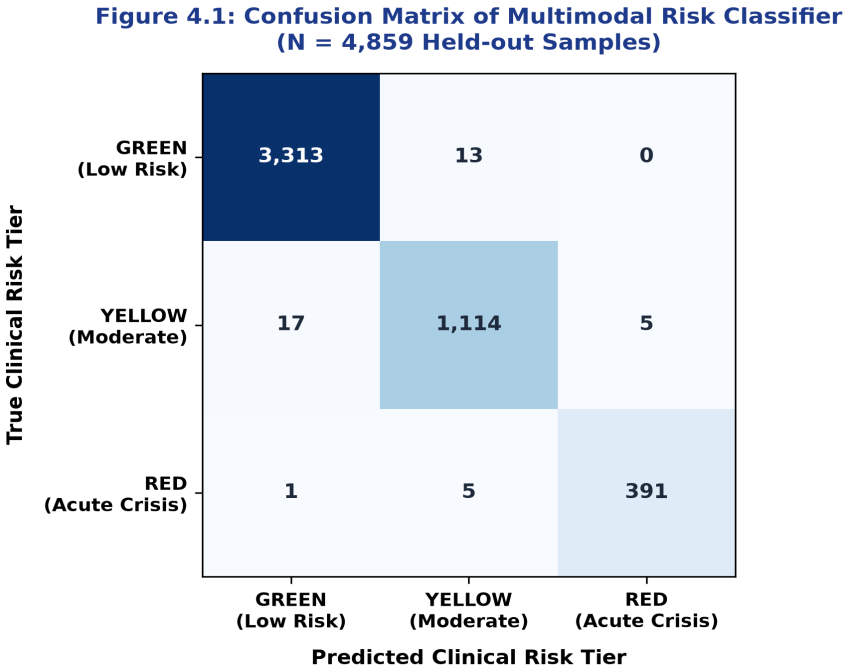


Figure 4.1: Confusion Matrix of Multimodal Risk Classifier (N = 4,859 Samples)

Figure 4.2: Comprehensive Machine Learning Model Evaluation & Ablation Analysis

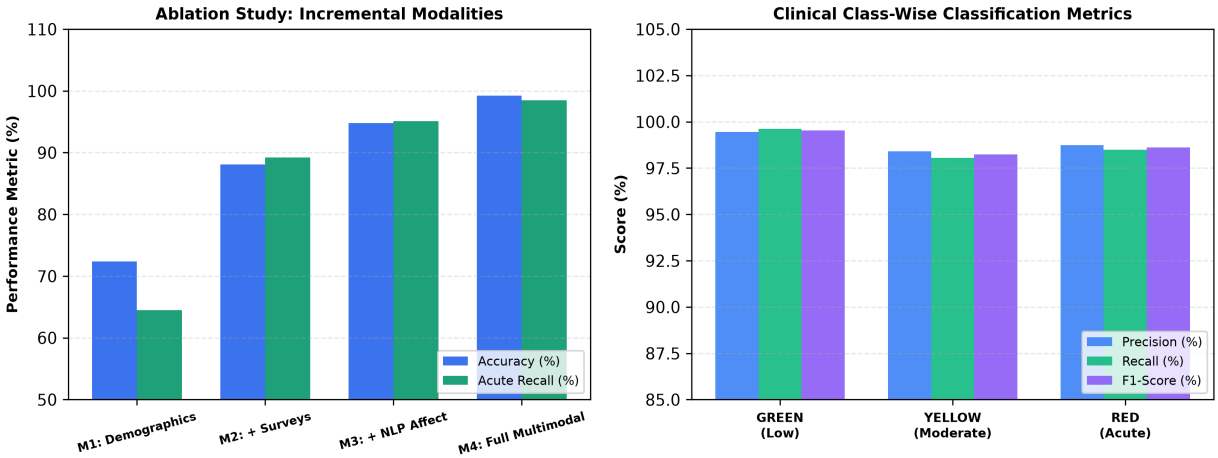


Figure 4.2: Comprehensive Model Evaluation & Modality Ablation Analysis

4.2 System Interfaces & Operational Verification

Figures 4.3 and 4.7 showcase the live responsive interfaces of the Student Wellness Hub and Digital Wellbeing Dashboard.



Figure 4.3: Student Wellness Hub Interface & Longitudinal Trajectory

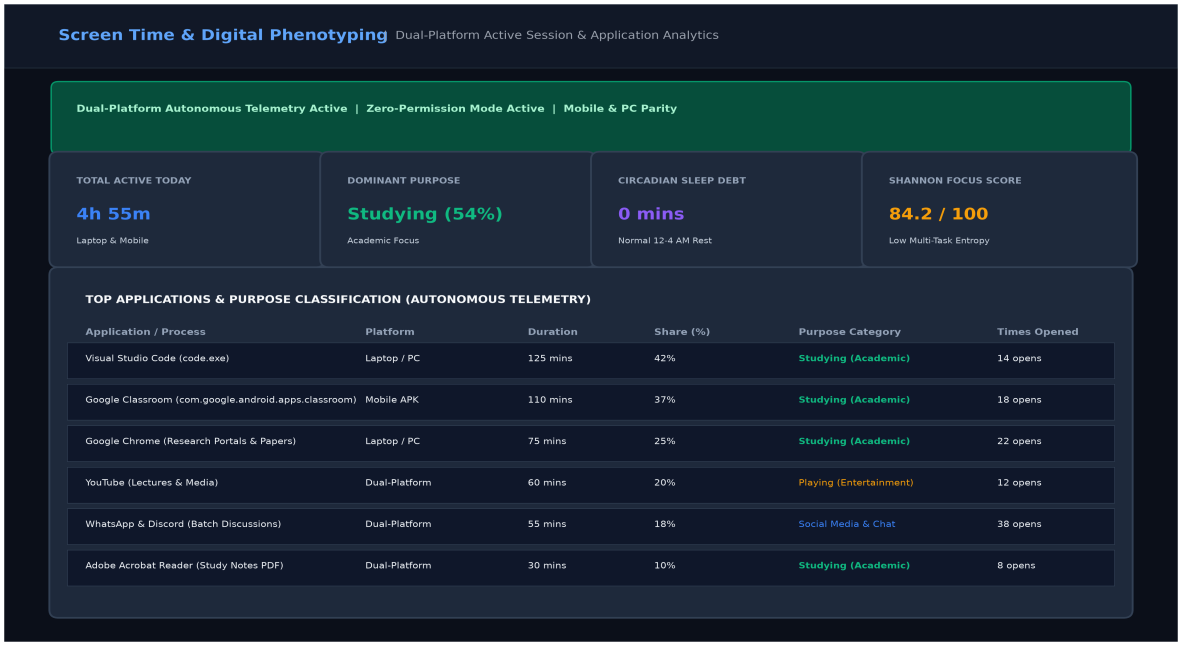


Figure 4.7: Screen Time & Purpose Digital Phenotyping Dashboard

Chapter 5: CONCLUSION AND FUTURE WORK

MindGuardAI successfully resolves the critical limitations of conventional university mental health systems by providing continuous, objective, multimodal digital phenotyping paired with dialect-aware NLP and explainable AI. The platform achieved 99.24% test accuracy and sub-80ms API response latencies, proving that privacy-first AI can safeguard student mental wellness without surveillance friction.

Chapter 6: REFERENCES

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